

HONGZHE XIE

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EDUCATION

University of California, Davis

Expected May 2028

Davis, CA

EXPERIENCE

Full-Stack Software Engineer

Oct 2025–Present

School of Engineering, University of California, Merced

Merced, CA

- Independently architected and built, and now operate, the official production platform for a program serving all campus use; a block-based Content Management System, delivered registration, barcode ticketing and check-in, and traceable administrative workflows with Django REST Framework and React/TypeScript.
- Developed an AI-assisted phishing detector combining SPF/DKIM/DMARC and sender-alignment validation, punycode, typosquat, and URL heuristics, plus optional Amazon Bedrock review with a rules-only fallback.
- Built a guarded dual-tier agentic workflow enabling admin management while providing safe, read-only public event queries, secured by strict RBAC, bounded context, IP-based token budgets, and anti-privilege escalation guardrails.
- Hardened authentication and delivery with rotating JWTs, RSA-encrypted password transport, one-time email/SMS challenges, endpoint throttling, CSP/CSRF/ HTS, CodeQL, Bandit, Trivy, Gitleaks and approval-gated deployments to AWS ECS Fargate and Amplify.

Computer Vision / AI Engineering Intern

Jul 2023–Oct 2023

China Academy of Building Research

Shanghai, China

- Developed a Python-based semantic-segmentation pipeline to identify deployable rooftop photovoltaic areas in aerial imagery of rural villages.
- Prepared and annotated image datasets, trained deep-learning models, and supported estimates of potential solar-energy generation from segmented rooftop areas.

Financial Investment Analysis Trainee

Jul 2023

Citigroup Inc.

Shanghai, China

- Selected for Citi's Financial Investment Analysis Training Program; completed intensive training in financial markets, banking operations, company fundamentals, and investment analysis.
- Applied Excel and Python to financial modeling and KPI analysis, translating company-performance data into concise dashboards and investor-oriented findings.

SELECTED PROJECTS

MobileID | [GitHub](#)

2025–2026

Python, Django, Vue, PostgreSQL, DynamoDB, Docker, GitHub Actions

- Developed a Django/Vue identity platform for PDF417 barcode generation, profile management, and device-aware sessions backed by PostgreSQL and DynamoDB.
- Implemented defense-in-depth authentication with HTTP-only cookie JWTs, CSRF validation, refresh-token rotation and blacklisting, immediate session revocation, Argon2 hashing, and scoped credential-stuffing controls.
- Enforced Bandit, Safety, Trivy, and dependency-audit gates alongside Django tests with a 70% coverage threshold, Playwright E2E tests, and Docker/Vite build validation.

TokenRouter | [GitHub](#)

2026

Go, React, OpenAI/Claude/Gemini APIs, GORM, Redis, Casbin, Docker

- Engineered an OpenAI-compatible LLM gateway with cross-provider protocol conversion, priority/weighted routing, failure-aware retries, and concurrency-safe reservation and settlement of token quotas.
- Protected relay and control planes with SSRF-safe outbound dialing, JWT refresh rotation and replay detection, Casbin RBAC, WebAuthn/TOTP, step-up verification, rate limits, and race-tested quota logic.

PROFESSIONAL DEVELOPMENT

Google Cloud Next 2026

Apr 2026

Technical Conference Attendee

AI Security & Agentic Systems

- Engaged in technical deep dives on enterprise agentic architectures, focused on multi-agent orchestration, Vertex AI Agent Builder, and AgentOps lifecycle management.
- Explored AI security, model governance, and runtime defense using Model Armor, Security Command Center (SCC) for GenAI, and Bedrock/Vertex-level guardrail frameworks.
- Evaluated modern GenAI data/compute infrastructures, including BigQuery vector search integration, Hypercomputer scaling (A3/TPU v5p), and secure Tool Calling/ADK patterns.

Google I/O 2025

May 2025

Developer Conference Attendee – Google

AI Agents & Web Engineering

- Attended keynote and developer sessions on Gemini, Android, web technologies, multimodal systems, and the architecture of AI-native applications.
- Participated in hands-on Google ADK workshops focused on structured tool calls, agent orchestration, modular task flows, and web-performance optimization.

Microsoft AI Day 2024

2024

Technical Program Participant – Microsoft

Enterprise & Responsible AI

- Engaged with Microsoft AI practitioners on emerging model capabilities, responsible deployment practices, and production use cases across enterprise environments.
- Studied integration tradeoffs involving security, governance, reliability, human oversight, and organizational adoption of AI-enabled systems.

TECHNICAL SKILLS

Languages: Python, C/C++, Go, TypeScript/JavaScript, SQL

AI & Agent Engineering: Google ADK, Gemini, Amazon Bedrock, agent orchestration, structured tool/API calling, multimodal pipelines, LLM application security

Application & Identity Security: API security, phishing detection, WebAuthn/FIDO2, TOTP/MFA, JWT/OAuth, RBAC/Casbin, CSRF/XSS/SSRF mitigation, rate limiting, audit logging

Security Testing & Tooling: CodeQL, Bandit, Trivy, Gitleaks, OSV, Safety, Postman, SAST, dependency, container, and secret scanning

Backend & Data: Django/DRF, Celery, Redis, PostgreSQL, MySQL, DynamoDB, GORM, REST APIs

Frontend & Testing: React, Vue.js, Vite, Playwright, Django testing, end-to-end testing

Cloud & DevOps: AWS ECS Fargate, Amplify, S3, SES, CloudWatch, GCP, Docker, Nginx, GitHub Actions, CI/CD, SSH